INCH-POUND

MIL-DTL-14803B
25 February 2005
SUPERSEDING
MIL-I-14803A
17 March 1992

DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, DELAY, M30A1, M32A1, M92, M43A1, and M120 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M30A1, M32A1, M43A1, M92, and M120 Delay Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS AND STANDARDS

MIL-A-2550	Ammunition, General Specification For
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520225	Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
8798142	Box, Packing, Ammunition, For Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
8798143	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114
9216933	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, Delay, M10, M30A1, M31, M32A1, M42, M43A1, M72, M92, or M114

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4	Sampling Procedures and Tables for Inspection by Attributes
ANSI/ASQ A8402	Quality Management and Quality Assurance - Vocabulary

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 5500 to 6500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

3.3 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge retainer, and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum (see 4.4.4.2).

3.4 Initiator pin security. The initiator pin of the assembled initiator, less cartridge retainer and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin (see 4.4.4.3).

3.5 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied (see 6.3.2).

3.6 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction (see 4.5.5).

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.7.1 Integrated plant products. When remanufacturing and loading of assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.7.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus Class 3A or 3B thread gage required on new hardware for initiators.

3.8 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4)

4.2 Quality assurance terms and definitions. Reference shall be made to ANSI/ASQ A8402 to define the quality assurance terms used.

4.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.3.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68 to 77°F
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psia

4.4 Quality conformance inspection.

4.4.1 Lot.

4.4.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation, and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.4.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 600 assemblies, maximum, unless otherwise specified by contract.

4.4.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

rejected.

4.4.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.4.2.2 Retainer, Cartridge, Drawing 8594860, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
1	Metering slots in base missing	Visual
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Elongated slots missing	Visual
104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish on internal radius and base improper	Visual
108	Burr	Visual
109	Foreign matter	Visual
Minor		
None defined		

4.4.2.3 Chamber, Initiator, Drawing 8594857, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of large thread, min	Gage
102	Minor diameter of large thread, min	Gage
103	Pitch diameter of small thread, min	Gage
104	Major diameter of small thread, min	Gage
105	Thread damaged	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish, end of gas port, improper	Visual
109	Burr	Visual
110	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Minor
None defined

4.4.2.4 Cap, Drawing 8595404, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
1	Metal defective in mounting flange surface	Visual
Major		
101	Pitch diameter of thread, max	Gage
102	Minor diameter of thread, max	Gage
103	Thread damaged	Visual
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
None defined		

4.4.2.5 Pin, Initiator, Drawing 8595403, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish on body and body groove improper	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.4.2.6 Housing, Firing Pin, Drawing 8595408, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Critical
None defined

Major		
101	Pitch diameter of internal thread, max	Gage
102	Minor diameter of internal thread, max	Gage
103	Pitch diameter of external thread, min	Gage
104	Major diameter of external thread, min	Gage
105	Thread damaged	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Surface finish of cavity chamfer improper	Visual
109	Burr	Visual
110	Foreign matter	Visual

Minor
None defined

4.4.2.7 Pin, Firing, Drawing 8595407, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish improper	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
None defined		

4.4.2.8 Pin, Safety, Drawing 8595405, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter of wire	Measure
102	Profile incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter	Visual

Minor
None defined

4.4.2.9 Spring, Initiator, Drawing 8595406, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Load at specified heights	Gage
102	Height of solid, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr or sharp edges	Visual
106	Foreign matter	Visual
Minor		
201	Diameter outside, closed, max	Gage
202	Diameter inside, min	Gage
203	Free length	Gage

4.4.2.10 Seal, Drawing 10524102, covering a detail of Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked, or split	Visual
105	Flash	Visual
106	Foreign matter	Visual
Minor		
201	Diameter outside, max	Gage
202	Width, max	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

203	Thickness, max	Gage

4.4.2.11 Assembly, Drawing 10520225, less Drawings 8595403, 8595405, 8594857, 8594860 and NAS 832-4.

Characteristics	Defect	Method of Inspection
Critical		
1	Protrusion of firing pin	Gage
2	O-ring missing	Visual
3	Steel ball missing	Visual
Major		
None defined		
Minor		
201	O-ring not lubricated	Visual

4.4.2.12 Assembly, Drawing 10520225.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Initiator pin not free to rotate	Manual
102	Safety pin loose or deformed	Manual/Visual
103	Any part missing	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish inadequate or defective	Visual

4.4.2.13 Box, Fiberboard (Inner) Packing Ammunition, Drawing 8798143.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.4.2.14 Box, Packing, Ammunition, Drawing 8798142.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.4.2.15 Box, Fiberboard (Outer) Alternative, Packing Ammunition, Drawing 9216933.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber assembly to comply with the requirement of 3.2 shall be cause for rejection of the subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

4.4.4.2 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.4.4.3 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

4.4.4.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.4.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.5. Failure of any unit in the inspection sample to comply with 3.6 shall be cause for rejection of the assembly.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to the application of protective finish the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with the requirements of 3.2.

4.5.2 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.5.3 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.5.4 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in the north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. Since the initiator was developed specifically for use in military aircraft it has no commercail application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. The responsible inspection activity (see 4.1)

d. Any special inspection conditions (see 4.3)

e. Level of packing and marking (see 5.1 and 6.5)

6.3 Definitions.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: cap and chamber.

6.4 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.

Life-saving systems, aircrew

6.5 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.5.1 Packing and marking shall be Level A or B, as specified by the contracting activity.

6.5.1.1 Level A. Level A shall be in accordance with Drawings 8798143 and 8798142 or SPI F00-138-3770.

6.5.1.2 Level B. Level B shall be in accordance with Drawings 8798143 and 9216933 or SPI F00-138-3770.

6.5.1.3 Lot numbers. Lot numbers shall be in accordance with MIL-STD-1168.

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I058.

Custodian:
Navy - OS
Air Force - 99
Review activity:
Air Force – 70

Preparing activity
Navy - OS
(Project 1377-2005-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:47Z
Check the source to verify that this is the current version before use.